CFPB CONSUMER COMPLAINTS ANALYSIS

1. Introduction

This project analyzes over 14 million consumer complaints from the Consumer Financial Protection Bureau (CFPB) database. The dataset captures real complaints filed by consumers against financial institutions in the United States.

The objective of this analysis is to understand:

- Why customers file complaints
- How complaint behavior differs across users
- What patterns exist in complaint resolution
- What customers are actually saying in complaint narratives

Rather than focusing only on operational metrics, this analysis combines behavioral segmentation and NLP techniques to uncover deeper patterns behind customer dissatisfaction.

2. Data Overview

The dataset includes the following key attributes:

- Product and sub-product categories
- Complaint issue and sub-issue
- Consumer complaint narratives (text data)
- Company responses
- State and submission channel
- Timeliness of response
- Consumer dispute status

The dataset is large-scale, messy, and includes:

- missing values
- unstructured text
- inconsistent labels and noise

This reflects real-world data conditions in financial complaint systems.

3. Data Preparation

Several cleaning and transformation steps were performed:

- Standardized product categories into higher-level groups
- Handled missing values strategically based on business meaning
- Created behavioral flags such as:
    - presence of complaint narrative
    - public response availability
    - timely response indicator
- Cleaned and preprocessed text data for NLP analysis
- Removed noise tokens and irrelevant legal placeholders from text

These steps ensured the dataset was suitable for both behavioral and text analysis.

4. Exploratory Data Analysis (EDA)

The initial analysis focused on identifying structural patterns in complaints.

Key Findings:

- Credit-related complaints dominate the dataset, far exceeding other product categories
- The most common issues are related to:
    - inaccurate credit reporting
    - improper use of credit reports
    - investigation failures
- A small number of large financial institutions and credit bureaus account for a significant share of complaints
- Complaint volumes have increased significantly over time, showing growing consumer engagement with financial dispute systems
- Despite high complaint volumes, response timeliness remains consistently high across all categories

5. Behavioral Segmentation

To better understand customer behavior, complaints were grouped into segments based on engagement and structure.

Segments Identified:

- High Engagement Complaints
- Low Engagement Complaints
- System-Driven Credit Complaints
- Other Complaints

Key Insights:

- The largest segment is System-Driven Credit Complaints, indicating widespread recurring issues in credit reporting systems
- Low Engagement complaints also represent a large portion of users, suggesting many complaints are submitted without detailed narratives
- High Engagement complaints, while smaller in volume, show more detailed customer involvement
- Response timeliness remains consistently high across all segments, suggesting operational efficiency in handling complaints

6. Text Analysis (NLP)

Natural Language Processing was used to analyze complaint narratives.

Key Findings:

- Complaint language is dominated by terms related to:

    - credit reports
    - account management
    - inaccurate information
    - debt and payments
    - dispute and investigation processes
- Word cloud analysis confirms strong clustering around financial reporting and dispute resolution terms
- Sentiment analysis shows a mix of emotional tones, though complaint language is heavily influenced by technical and financial terminology rather than pure emotional expression

7. Key Insights

Across all layers of analysis, structured data, segmentation, and text analysis. Consistent patterns emerge:

- Credit reporting accuracy is the dominant driver of complaints
- Investigation and dispute handling processes are central points of failure
- Customer behavior varies in engagement level, but complaint themes remain consistent
- Operational response speed is high, but does not necessarily eliminate recurring complaint types
- Complaint narratives reflect issues of trust, accuracy, and system reliability rather than isolated service dissatisfaction

8. Conclusion

This analysis demonstrates that consumer complaints in the financial system are largely driven by recurring structural issues in credit reporting and dispute resolution processes.

While companies respond quickly to complaints, the persistence of similar issues across millions of records suggests deeper systemic challenges rather than isolated customer service failures.

The combination of behavioral segmentation and text analysis provides a clearer understanding of how customers experience financial systems and where dissatisfaction is concentrated.